



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY

LONERE- RAIGAD – 402103

Winter Semester Examination – Dec. – 2019

Branch – B. Pharmacy Third Year

Subject – Pharmacognosy and Phytochemistry II

Marks- 75

Date-23/12/2019

Semester- V

Subject Code- BP504T

Time- 03 Hrs

- Instructions- i) All questions are compulsory
ii) Figures to the right indicate full marks
iii) Draw the diagrams or flow charts wherever necessary

Q. No.1 Multiple Choice Questions (MCQs) (Answer all the questions) = $20 \times 1 = 20$

A] Shikimic acid pathway is an important intermediate pathway from carbohydrate for the biosynthesis of unit.

- a. C_5-C_3 b. $C_{12}-C_6$ c. L-Ornithin d. No any one

B] ^{14}C have an half-life of years is usefully in biological research.

- a. 8.8 b. 20 c. 5,700 d. 9,700

C] Chhotachand is in activity.

- a. Antineoplastic b. Antihypertensive c. Antidiabetic d. Antiemetic

D] Preparation of green tea, the leaves are shaken out on the pans.

- a. Aluminium b. Sliver c. Gold d. Copper

E] Allied species of *Dioscorea*

- a. *Dioscorea flouribunda* b. *Dioscorea villosa* c. *Dioscorea speciosus* d. all of these

F] Eugenol constituent is present in

- a. Fennel b. Catechu c. Clove d. Opium

G] Goldbeater's skin test is used for detection of

- a. Tannins b. Alkaloids c. Glycosides d. Both b and c

H] The benzoic acid contents in Sumatra benzoin is not less than %.

- a. 12 b. 60 c. 76 d. 90

I] Senna leaf represent histological characters is

- a. Dorsiventral b. Cassia c. Isobilateral d. all of these

Page 1/3

- J] Gentian root present major glycosides.
 a. Coumarin b. Isothiocyanate c. Sterol d. Bitter
- K] The content of artemisin is increased by 30% by applying growth regulator such as before harvesting.
 a. Chloromequat b. Potassium c. Thiamine d. Gibberellin
- L] Liquorice should have a water soluble extractive not less than % w/w.
 a. 02 b. 90 c. 10 d. 20
- M] Each ml of 0.1 M perchloric acid equivalent to gm. of quinidine.
 a. 1.02490 b. 2.02490 c. 0.02490 d. No any one
- N] Curcumin and other related curcuminoids are reported to be responsible for the colour.
 a. Green b. Yellow c. Red d. Colourless
- O] The alkaloids containing phytoconstituents give reddish brown ppt. with reagent.
 a. Dragendorff's b. Mayer's c. Hager's d. Tannic acid
- P] Taxus genus also called as
 a. Yam b. Yews c. Mulethi d. Tea
- Q] is used for purification of materials present in a crude extract.
 a. Percolation b. Maceration c. Fractional distillation d. Sublimation
- R] Spectroscopy is used for detection of functional groups in biomolecules.
 a. Mass b. N.M.R. c. I.R. d. U.V.
- S] U.V. and Visible spectroscopy measurement of light absorption by substances in wavelength region from
 a. 190-900 b. 190-380 c. 380-900 d. 1-90
- T] Electrophoresis is an process which separates charged particles in a fluid using a field of electrical charge.
 a. Spectroscopic b. Chromatographic c. Fermentation d. Electrokinetic

Q. No.2 Long Answers (Answers 2 out of 3) =

2×10 = 20

- A] Explain Shikimic acid pathway.
- B] Write biological source, family, chemical constituents, uses of;
i. Vinca ii. Tea iii. Senna iv. Dioscorea v. Myrrh.
- C] Explain Soxhlet and Supercritical fluid extraction and its Uses.

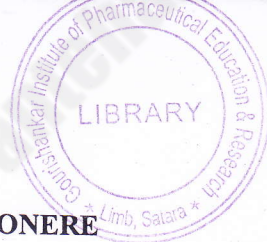
Q.No.3 Short Answers (Answers 7 out of 9) =

7×5 = 35

- A] Write a note on radioactive isotopes.
- B] Explain Clove and Catechu.
- C] Isolation, Identification and Analysis of Rutin.
- D] Give short note on Industrial production, estimation and utilization of atropine.
- E] Define spectroscopy and chromatography and write its application.
- F] Define alkaloid, terpenoids, and explain its chemical tests.
- G] Draw T.S. of fennel fruit and explain in brief it.
- H] Write a note on Benzoin and difference between Sumatra benzoin and Siam benzoin.
- I] Explain in brief Opium and give note on preparation of opium.

----- END OF PAPER -----

Page 3/3



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
End Semester Examination – Winter 2022

Date: 15/02/2023

Course: B. Pharmacy
Subject Name: Pharmacognosy & Phytochemistry-II
Max Marks: 75

Sem: V
Subject Code: BP504T
Duration: 3 Hr.

I. Objective Type Questions

(10 X 2 = 20)

- i) Define the term extraction and electrophoresis.
- ii) Explain morphology of Senna leaflet
- iii) Draw a well labelled diagram of Soxhlet assembly and percolator
- iv) Define the term metabolic pathway and metabolites.
- v) Write the biological source and identification test of Cinchona bark.
- vi) Draw a well labelled diagram of Fennel fruit transverse section.
- vii) Define radioactive isotopes with examples.
- viii) Write Biological source and uses of Asafoetida.
- ix) Classify tannins with suitable examples
- x) Write in short applications of HPLC.

II. Long Answers (Answer 2 out of 3)

(2 X 10 = 20)

- i) Define and classify alkaloids with suitable examples and write in detail about Vinca.
- ii) Discuss in detail methods of extraction of Phytoconstituents.
- iii) Explain in detail shikimic acid pathway with its role in biosynthesis of metabolites.

III. Short Answers (Answer 7 out of 9)

(7 X 5 = 35)

- i) Write synonyms, biological source, chemical constituent, identification test and uses of Digitalis.
- ii) Write Industrial production, estimation and utilization of Sennosides.
- iii) Briefly explain Steps involved in tracer techniques.
- iv) Write about isolation, identification, analysis of Curcumin
- v) Define the term chromatography and add a note on TLC
- vi) Write Industrial production, estimation and utilization of Atropine
- vii) Write difference between Sumatra benzoin & Siam benzoin.
- viii) Give isolation identification and analysis of Caffeine.
- ix) Write synonyms, biological source, chemical constituent, morphology and uses of Fennel.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
End Semester Examination – SUMMER 2023

Date: 26/07/2023

Course: B. Pharmacy

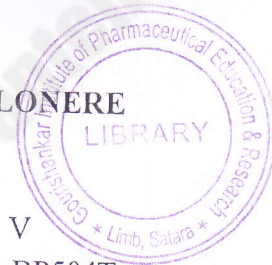
Sem: V

Subject Name: Pharmacognosy & Phytochemistry-II

Subject Code: BP504T

Max Marks: 75

Duration: 3 Hr.



I. Objective type Questions

(10 X 2 = 20)

- i. Define extraction and Marc.
- ii. What are resins and terpenoids?
- iii. Give the source and utilization of Taxol.
- iv. Define metabolic pathway with an example.
- v. Give the source and identification of Quinine.
- vi. Write morphology of Coriander.
- vii. Explain radioactive isotopes with examples.
- viii. Write Biological source and uses of Asafoetida.
- ix. Write utilization of Vincristine and vinblastine.
- x. Give types and applications of paper chromatography.

II. Long Answers (Answer 2 out of 3).

(2 X 10 = 20)

- i. What are Alkaloids? Classify them with examples and write in detail about Vinca.
- ii. Write in brief various methods of extraction of Phytoconstituents.
- iii. Explain Shikimic acid pathway in detail.

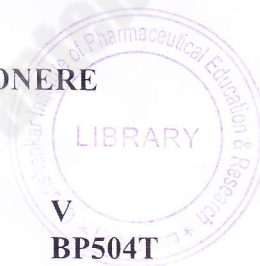
III. Short Answers (Answer 7 out of 9)

(7 X 5 = 35)

- i. Write synonyms, biological source, chemical constituent, identification test and uses of Digitalis.
- ii. Write Industrial production, estimation and utilization of Sennosides.
- iii. Briefly explain Steps involved in tracer techniques.
- iv. Write about isolation, identification, analysis of Curcumin
- v. Define chromatography and make a note on TLC
- vi. Write Industrial production, estimation and utilization of Atropine
- vii. Write difference between Sumatra benzoin & Siam benzoin.
- viii. Give isolation identification, analysis of Caffeine.
- ix. Write synonyms, biological source, chemical constituent, morphology and uses of Fennel.

Course: T.Y.B. Pharmacy
Subject Name: Pharmacognosy and Phytochemistry - II
Max Marks: 75

Sem: V
Subject Code: BP504T
Duration: 3 Hrs.

**Instructions:**

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

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- Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20**
- i) Write identification test of Quinine.
 - ii) Define metabolites and metabolic pathway.
 - iii) Suggest a chemical test for tropane alkaloids and explain it.
 - iv) Draw a neat labelled diagram of the T. S. of Ephedra stem.
 - v) Write the biological source and uses of Senna leaf.
 - vi) Give the source and utilization of Artemisinin.
 - vii) Write adulterants of Clove flower bud.
 - viii) Write the chemical structure of Reserpine and Rutin.
 - ix) Define radioactive isotopes with examples.
 - x) Name any two resinous drugs and give their uses.
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- Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20**
- i) Explain Shikimic acid pathway and its significance.
 - ii) Enumerate any four modern methods of extraction and explain in detail about Super Critical fluid extraction with their advantages and disadvantages.
 - iii) Discuss the industrial production and utilization of Atropine and caffeine.
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- Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35**
- i) Write the isolation and analysis of Glycyrrhetic acid.
 - ii) Give the biological source, chemical tests and uses of Benzoin.
 - iii) Explain column chromatography.
 - iv) Prepare a note on Spectroscopy in identification of Phytoconstituents.
 - v) Write botanical source and chemical constituents and uses of Digitalis.
 - vi) Define Glycoside. Classify glycosides based on the glycosidic linkages.
 - vii) Compare and contrast Pale catechu and Black catechu.
 - viii) Write biological source, chemical constituents and uses of Deadly night shade plant.
 - ix) Define tracer technique and explain in brief steps involved in the tracer techniques.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: V

Subject Name: Pharmacognosy & Phytochemistry-II

Subject Code: BP504T

Max Marks: 75

Date: 21-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams/ figures wherever necessary
3. Figures to right indicate full marks

Q.1.	Objective Type Questions (Answer All the Questions)	(10X2) =20
i)	Define phytochemistry.	
ii)	Define Balsams with examples.	
iii)	Give identification tests for Saponin glycosides.	
iv)	Explain the isolation method of Menthol.	
v)	Write the medicinal uses of Vincristine and Vinblastine.	
vi)	Write source ,chemical constituents and uses of Fennel fruit.	
vii)	Enlist methods of extraction.	
viii)	Give source and chemical constituents of Licorice root.	
ix)	Define primary and secondary metabolites.	
x)	Write the uses of Diosgenin.	
Q.2.	Long Answers (Answer 2 out of 3)	(10X2) =20
i)	Discuss the Industrial production, estimation and utilization of Caffeine and Artemisinin.	
ii)	Define Alkaloids, explain its types and write the pharmacognostic account on Opium.	
iii)	Elaborate and draw the Shikmic acid pathway.	
Q.3.	Short Answers (Answer 7 out of 9)	(5 X 7) = 35
i)	Discuss the Isolation, Identification and Analysis of Atropine.	
ii)	Explain the drug Senna.	
iii)	Define volatile oils and discuss the drug Cinnamon bark.	
iv)	Elaborate the applications of Chromatography in phytochemistry.	
v)	Write a note on Modern methods of extraction.	
vi)	Explain the identification tests for Asafoetida and Myrrh.	
vii)	Discuss the Industrial production and utilization of Digoxin.	
viii)	Write source, chemistry and uses of Belladonna and Aloe.	
ix)	Discuss the Isolation, Identification and Analysis of Curcumin.	
*** END OF THE PAPER ***		

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